Tutorium 12

P12.1:

$U_N = \mathbb{R} \times \mathbb{R}^+$

$U_E = \mathbb{R}^+ \times \mathbb{R}$

$\varphi_N(x,y) = x$

$\varphi_E(x,y) = y$

$\boxed{\varphi_E \circ \varphi_N^{-1}}$

$S^1 = \{(x,y) \in \mathbb{R}^2 \mid x^2 + y^2 = 1\}$

$\varphi_N(U_N \cap S^1 \cap U_E) = (0,1) =: V_N$

$\varphi_E(U_E \cap S^1 \cap U_N) = (0,1) =: V_E$

$\varphi_E \circ \varphi_N^{-1} : V_N \longrightarrow V_E$

$\varphi_E \circ \varphi_N^{-1}(\underbrace{x}_{\in V_N}) = \varphi_E(x, \sqrt{1-x^2}) = \sqrt{1-x^2} \in (0,1) = V_E$

$\varphi_N \circ \varphi_E^{-1} : V_E \longrightarrow V_N, \quad y \longmapsto \sqrt{1-y^2}$

P12.2: $f: \mathbb{R}^4 \to \mathbb{R}^2$

$(x_1, x_2, x_3, x_4) \longmapsto \begin{pmatrix} f_1(x) \\ f_2(x) \end{pmatrix} = \begin{pmatrix} x_1 + x_4^3 \\ x_1 x_4 - x_2^2 - x_3^3 \end{pmatrix}$

(a) z.z.: $M = f^{-1}(\{(0,-1)\})$ eine UMF $\mathbb{R}^4$

$\downarrow$ r.W.

Bew.: $f \in C^\infty(\mathbb{R}^4, \mathbb{R}^2)$, $y = (0,-1) \in \mathbb{R}^2$

$(0,0,1,0) \in M \Rightarrow M \neq \emptyset$

$$J_f(x) = \begin{bmatrix} 1 & 0 & 0 & 3x_4^2 \\ x_4 & -2x_2 & -3x_3^2 & x_1 \end{bmatrix} \begin{matrix} \mathrm{I} \\ \mathrm{II} \end{matrix}$$

$\mathrm{II} = \mathrm{II} - x_4 \mathrm{I} \rightsquigarrow \begin{bmatrix} 1 & 0 & 0 & 3x_4^2 \\ 0 & -2x_2 & -3x_3^2 & x_1 - 3x_4^2 \end{bmatrix}$

$\nexists x \in M : x_2 = x_3 = 0 \wedge x_1 = 3x_4^3$

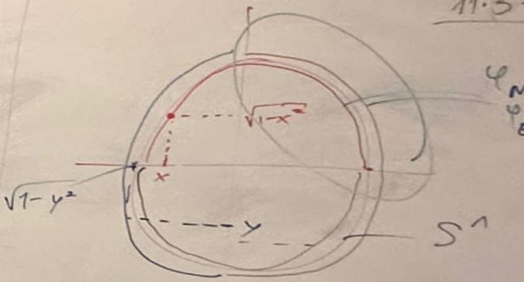
Widerspruchsbew.: $\exists x \in M$:

$x_1 + x_4^3 = 0 \quad \wedge \quad x_1 x_4 - x_2^2 - x_3^3 = -1$

$3x_4^3 + x_4^3 = 4x_4^3 = 0 \Rightarrow x_4 = 0 \Rightarrow x_1 = 0$

$0 = -1$ ↯

$11 \cdot 3 + 6 \quad \% \, 70$



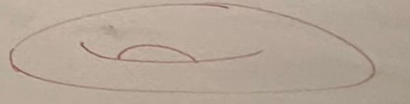
φ_{NORD}

φ_{EAST}

M UMF – 2 Karten: $\varphi_1 : U_1 \cap M \to \mathbb{R}^n$

$\varphi_2 : U_2 \cap M \to \mathbb{R}^n$

$\varphi_2 \circ \varphi_1^{-1} : \varphi_1(U_1 \cap U_2 \cap M) \longrightarrow \varphi_2(U_1 \cap U_2 \cap M)$



$\hookrightarrow \forall x \in M: \operatorname{rang} \mathcal{J}_f(x) = 2$

M ist eine UMF des $\mathbb{R}^4$ mit $\dim M = 4 - 2 = \underline{2}$

(b) $p = (-1, 0, 0, 1)$

$$\mathcal{J}_f(p) = \begin{bmatrix} 1 & 0 & 0 & 3 \\ 1 & 0 & 0 & -1 \end{bmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} x_1 + 3x_4 \\ x_1 - x_4 \end{pmatrix} \overset{!}{=} \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\Rightarrow x_1 = x_4 = 0$$

$$T_p M = \operatorname{Ker} \mathcal{J}_f(p)$$

$$= \{ y \in \mathbb{R}^4 \mid y_1 + 3y_4 = 0 \wedge y_1 - y_4 = 0 \}$$

$$= \{ y \in \mathbb{R}^4 \mid y_1 = y_4 = 0 \} = \operatorname{span}\left\{ \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} \right\}$$

$$T = \left((R + r\cos v)\cos u, (R + r\cos v)\sin u, r\sin v \right), \quad u, v \in [0, 2\pi]$$

P12.3: $F: \mathbb{R}^4 \to \mathbb{R}^2$, $F(x) = \begin{pmatrix} x_1^2 + x_2^2 - 1 \\ x_3^2 + x_4^2 - 1 \end{pmatrix}$, $\Pi := F^{-1}(\{(0,0)\})$

(a) $\underline{\text{z.z.}}$: Π eine 2-dim UMF von $\mathbb{R}^4$ ist.

$F \in \mathcal{C}^\infty(\mathbb{R}^4, \mathbb{R}^2)$, $y = (0,0) \in \mathbb{R}^2$, $(1,0,1,0) \in \Pi \Rightarrow \Pi \neq \emptyset$

$$\mathcal{J}_F(x) = \begin{pmatrix} 2x_1 & 2x_2 & 0 & 0 \\ 0 & 0 & 2x_3 & 2x_4 \end{pmatrix}$$

$4 - 2 = 2$

$x_1 = x_2 = 0 \vee x_3 = x_4 = 0$

$\Rightarrow \forall x \in \Pi$: x regulär $\Rightarrow \Pi$ eine 2-dim UMF von $\mathbb{R}^4$

(b) Atlas von $\mathbb{T}$

$$S^1_r := \{(x,y)\in\mathbb{R}^2 \mid x^2+y^2=r^2\}$$

analytisch: $\mathbb{T} = S^1_R \times S^1_r$

top.: $\mathbb{T} = S^1 \times S^1$

$\varphi\in[0,2\pi)$: $e^{i\varphi} \in \mathbb{C}_1 \cong \mathbb{R}^2$

$$\mathbb{T} = S^1\times S^1 : (e^{i\varphi_1}, e^{i\varphi_2}) \in \mathbb{C}_1^2 \cong \mathbb{R}^4$$

den $\mathbb{T}$ vollständig besch.:

$$\Phi : \mathbb{R}^2 \to \mathbb{T} \subseteq \mathbb{R}^4 : (\varphi_1, \varphi_2) \mapsto \begin{pmatrix}\cos\varphi_1\\ \sin\varphi_1\\ \cos\varphi_2\\ \sin\varphi_2\end{pmatrix}$$

(c) $N_p\mathbb{T} = \mathrm{span}\{\nabla F_1(p), \nabla F_2(p)\}$

$$= \mathrm{span}\left\{\begin{pmatrix}p_1\\ p_2\\ 0\\ 0\end{pmatrix}, \begin{pmatrix}0\\ 0\\ p_3\\ p_4\end{pmatrix}\right\}$$

$$J_\Phi(\varphi_1,\varphi_2) = \begin{pmatrix}-\sin\varphi_1 & 0\\ \cos\varphi_1 & 0\\ 0 & -\sin\varphi_2\\ 0 & \cos\varphi_2\end{pmatrix} \Rightarrow \operatorname{rang} J_\Phi(\varphi_1,\varphi_2) = 2$$

$$\Phi(0,0) = \Phi(2\pi,0) = (1,0,1,0)$$

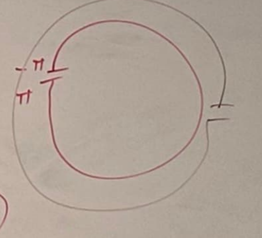
$$p = \Phi(\psi_1,\psi_2) \in \mathbb{T} \Rightarrow R_p = (\psi_1-\pi, \psi_1+\pi)\times(\psi_2-\pi, \psi_2+\pi)$$

$$\Phi_p := \Phi|_{R_p} \Rightarrow \Phi_p \text{ umkehrbar} \Rightarrow \Phi \text{ Diffeo}$$

$$U_p := \Phi_p(R_p) \subset \mathbb{T}$$ offen. ($U_p \subset \mathbb{T}$)

$$\varphi_p := \Phi_p^{-1} : U_p \to R_p$$

$$\mathcal{A} = \{(U_p, \varphi_p : U_p \to R_p) \mid p \in \mathbb{T}\}$$



$$T_p\mathbb{T} = (N_p\mathbb{T})^\perp = \mathrm{span}\left\{\begin{pmatrix}-p_2\\ p_1\\ 0\\ 0\end{pmatrix}, \begin{pmatrix}0\\ 0\\ -p_4\\ p_3\end{pmatrix}\right\}$$